Machine Markets: Economic-Contract Repairs and Exact Counterexamples

Research revision evidence

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1 Conditional Ambiguity Pricing and Allocation Surplus

Fix a one-unit ticket and a CARA dealer with utility $-\exp(-\gamma\pi)$, $\gamma > 0$. Under a candidate model θ, the ticket's monetary adverse mark is $Z_\theta \sim N(\mu_\theta, \sigma_\theta^2)$. A total spread receipt s produces profit $s - Z_\theta$. All costs below are monetary per this fixed ticket; other ticket quantities require the corresponding exposure scaling. Set

$$c_\theta = \mu_\theta + \frac{\gamma\sigma_\theta^2}{2}, \qquad c_2 = \sup_{\theta\in\Theta} c_\theta < \infty.$$

Assume $c_\theta \geq 0$ and nonnegative quotes. Competition imposes zero robust certainty equivalent, with no additional fixed costs or inventory obligations in this ticket experiment. The informed benchmark knows $\theta_0 \in \Theta$ and has $c_1 = c_{\theta_0}$.

Proposition 1 (Conditional robust quote and zero-premium cases)**.** *The minimum robust break-even quote is $s_2 = c_2$, while $s_1 = c_1$. Thus $\Delta = s_2 - s_1 \geq 0$, with strict inequality if and only if $\sup_\theta c_\theta > c_{\theta_0}$. Parameter nonidentification alone does not imply a positive premium, nor proportionality to a coordinate-dependent parameter diameter.*

Proof. The Gaussian moment-generating function gives

$$-\gamma^{-1}\log \mathbb{E}_\theta \exp[-\gamma(s - Z_\theta)] = s - \mu_\theta - \gamma\sigma_\theta^2/2 = s - c_\theta.$$

Its infimum over models is $s - c_2$. A nonnegative robust certainty equivalent therefore requires $s \geq c_2$. The informed calculation substitutes θ_0. If every observationally equivalent latent decomposition has the same relevant cost, $c_\theta = c_1$, then $\Delta = 0$ despite nonidentification. Rescaling coordinates of θ changes a Euclidean parameter diameter without changing any c_θ or quote. □

For an additional declared allocation-surplus criterion, let inverse demand be $v(V) = (V_0 - V)/b$, $b > 0$, and fixed effective marginal cost be c. Both volumes must be interior: $0 \leq V_0 - b(s_0 + \Delta) \leq V_0 - bs_0$. Treating a risk charge as this cost is an extra welfare assumption; transfers and risk sharing can change the relevant social criterion.

Proposition 2 (Allocation loss with an existing spread distortion)**.** *The surplus lost when the quote rises from s_0 to $s_0 + \Delta$ is*

$$\int_{V_0 - b(s_0+\Delta)}^{V_0 - bs_0} [v(V) - c], dV = b(s_0 - c)\Delta + \frac{b}{2}\Delta^2.$$

The triangle $b\Delta^2/2$ requires the efficient baseline $s_0 = c$. The result does not by itself identify an empirical welfare cost of a cohort-equivalence class.

Proof. Substitute $v(V) = (V_0 - V)/b$ and integrate the affine integrand over an interval of length $b\Delta$. Its endpoint values are $s_0 - c$ and $s_0 + \Delta - c$; interval length times their average gives the formula. □

2 An Explicit Capacity Choice and Its Stability Conditions

Let $A = E - \mu|I|P$ be available monetary capital, with positive mark P and inventory I in quantity units. A static capacity illustration fixes net execution edge $r \geq 0$, holding-cost coefficient $h > 0$, capital per posted quantity $m > 0$, and quantity bounds $0 \leq L_{\min} \leq L_{\max}$. Its problem is

$$\max_{L \in [L_{\min}, L_{\max}]} \{rL - hL^2/2\}, \qquad mL \leq A.$$

Here r has monetary/quantity units, h monetary/quantity-squared units, and m monetary/quantity units. Demand, fills and inventory costs are fixed primitives of this one-period illustration; it is not the solution to an unspecified dynamic competitive-dealer problem.

Proposition 3 (Capacity choice and quoting infeasibility)**.** *If $\bar{L} = \min(L_{\max}, A/m) < L_{\min}$, the quoting commitment is infeasible. Otherwise the unique optimum is*

$$L^* = \min\{\bar{L}, \max(L_{\min}, r/h)\}.$$

For a declared linear book spanning a price width $w > 0$, slope depth is $D = L^*/w$ and impact is $\kappa = w/L^*$ when $L^* > 0$. Increasing capital weakly raises capacity; the capacity is unchanged when its capital cap is nonbinding.

Proof. The objective is strictly concave and has derivative $r - hL$. Its unrestricted maximizer is r/h. Projection onto the nonempty feasible interval gives the unique solution. Its upper endpoint weakly increases with A, proving the comparative static. The price-impact identity is a declared linear clearing schedule, not inferred from the optimization alone. □

With fixed anchor response $\varphi > 0$, the depth-dependent price coefficient is $a = 1 - \kappa\varphi$. For one cohort with $0 < \eta < 1$ and weight q, the transition is

$$J(\kappa) = \begin{pmatrix} 1 - \kappa\varphi & \kappa q \\ \eta(1 - \kappa\varphi) & 1 - \eta + \eta\kappa q \end{pmatrix}.$$

Proposition 4 (Exact single-cohort depth stability boundary)**.** *For $\kappa > 0$, $\varphi > 0$ and $0 < \eta < 1$, $J(\kappa)$ is Schur stable precisely when*

$$q < \varphi, \qquad \kappa < \frac{4 - 2\eta}{(2 - \eta)\varphi - \eta q}.$$

If the earlier restriction $0 \leq a < 1$ is maintained, it additionally requires $0 < \kappa\varphi \leq 1$. Higher impact can therefore move eigenvalues as well as transient gains. It does not necessarily create large non-normal amplification.

Proof. Let $T = 2 - \eta + \kappa(\eta q - \varphi)$ and $d = (1-\eta)(1-\kappa\varphi)$ be trace and determinant. The three strict Jury inequalities are

$$1 - T + d = \kappa\eta(\varphi - q) > 0,$$
$$1 + T + d = 4 - 2\eta - \kappa[(2-\eta)\varphi - \eta q] > 0,$$
$$1 - d = \eta + (1-\eta)\kappa\varphi > 0.$$

The first requires $q < \varphi$. This makes the denominator in the second bound positive, while the third is automatic. □

An exact positive-feedback counterexample uses $\varphi = 1$, $q = 1/10$, $\eta = 9/10$. At $\kappa = 1/10$,

$$J = \begin{pmatrix} 9/10 & 1/100 \\ 81/100 & 109/1000 \end{pmatrix}, \quad \|Je_1\|_2^2 = 14661/10000 > 1.$$

At the larger impact $\kappa = 1$,

$$J = \begin{pmatrix} 0 & 1/10 \\ 0 & 19/100 \end{pmatrix}, \quad \|J\|_2^2 = 461/10000 < 1.$$

Both are stable by the preceding inequalities. In the fixed Euclidean all-state experiment, the lower-impact model has $M \geq \sqrt{14661}/100 > 1$, whereas the higher-impact model has $M = 1$, since every positive-time power contracts and $J^0 = I$. Thus a universal depth-thinning/transient-danger implication is false even with positive feedback. This does not invalidate amplification results under their own fixed-spectrum and output assumptions.

3 Exact Cash, Inventory and Marked Equity

Let signed aggressive quantity $q_t > 0$ mean the dealer sells, with the trade occurring at the current midpoint P_t plus signed half-spread. Let fees f_t and funding charges g_t be monetary. With current inventory I_t and cash C_t,

$$I_{t+1} = I_t - q_t, \qquad C_{t+1} = C_t + q_t P_t + s_t|q_t| - f_t - g_t.$$

Mark equity after the subsequent price change at P_{t+1}.

Proposition 5 (Self-financing marked-equity identity)**.** *For $E_t = C_t + I_t P_t$,*

$$E_{t+1} - E_t = s_t|q_t| + (I_t - q_t)(P_{t+1} - P_t) - f_t - g_t.$$

The old expression $s_t|q_t| - I_t(P_{t+1} - P_t)$ is incompatible with this signed trade and inventory convention.

Proof. Substitute the cash and inventory updates into $E_{t+1} = C_{t+1} + I_{t+1}P_{t+1}$ and cancel $q_t P_t$ against the inventory trade. The resulting mark exposure is the post-trade inventory. A long inventory with no trade gains when the price rises. □

A fixed model loss budget B in monetary units is distinct from posted quantity and a norm radius in latent-state coordinates. Choosing $0 < B \leq A_0$ does not ensure future margin feasibility when positions or prices change. Conditional transient/input bounds require a financially specified output operator, and actual execution additionally requires fees, funding, currency conversion, margin rules and fundamental-price risk. These are calibration and information gates, not consequences of an algebraic eigenvalue test.